ACS-3922 Fall 2026

Assignment 2

Due Date: Monday, November 2nd at 11:59 pm

Note, you can use the assets from the course tutorials, but will also need to find some additional assets. Either you can design your own, or you can explore free assets available on OpenGameArt.org or kenney.nl. It is recommended to use downloaded assets at first, and if there is time and you are interested you can then create your own. Be sure to cite your sources in an MD file in your submission.

Part A – Platformer Game

In this assignment, you will create a single level of a platformer game. Create a new project and set up applicable folders for your different assets.

1. Use a single scene, named `MainScene`. You can use the assets provided with Tutorial 3 to start, and/or download/design your own as needed.

2. Create a 2D platformer level with the following requirements:
    - A steady follow-cam (size 3.5).
    - A world/level that:
        - Extends by at least 8 times the camera width.
        - Has at least two pits that require jumping over.
            - A life is lost if the player falls down a pit. *Hint: don't put anything below a pit and simply do a check to see if the player is below a certain y boundary value.*
        - Has at least two moving platforms that require jumping onto. These could be included with the pits or as a means to reach pickups (see below).
        - Includes at least two types of enemies.
            - One that moves horizontally and one that moves vertically.
                - If the player collides with horizontal moving enemy anywhere except the top of the enemy they lose a life. *Hint: look at the Try it Yourself in Tutorial 3 that discusses how to specifically detect a collision with the top of an object.*
                - If the player collides with a vertical moving enemy anywhere they lose two lives.
        - Includes at least two types of pickups:
            - One type must have at least 20 instances (e.g. coin); these do not have to be instantiated at run time and can be placed in the scene. This pickup should add 1 point to the player's score when picked up by the player.

        - One that is an animated object with at least 3 instances (e.g., a star); these do not have to be instantiated at run time and can be placed in the scene. This pickup should add 5 points to the player's score when picked up by the player. These should be harder to get to/more hidden than the previous pickup (e.g., coins).
            - Animation could be a spin, scaling, small vertical up/down movement, etc.
    - Includes the following checkpoints:
        - A checkpoint (e.g., a flag asset) about halfway through the level. The player will respawn here if they lose a life after the checkpoint, otherwise they will start at the beginning again.
        - A final game win checkpoint (e.g., a flag asset) at the end of the level. The player will win the game if they reach this checkpoint.
- A UI that:
    - Includes the current score (i.e., coins/stars picked up).
    - Includes the current lives count.
        - Note, at the beginning, the player should start with three lives.
    - Includes game messaging:
        - When the player loses their last life, a message should appear that says they lost the game/level. When a player reaches the final game win checkpoint, a message should appear that says they won the game/level.
        - For either a win or lose, you do not have to offer a way to restart or quit. You can simply rerun the game in the editor to restart.
            - However, you should remove the player from the game upon the displaying of either outcome and corresponding message.

Note, you must use inheritance for commonalities of your enemies and pickups. Please also read Part B now, as it will impact some game design you decide upon.

When working on the assignment it is very easy to spend lots of time on the design! Focus first on base functionality and requirements, and then if you have time you can polish aspects as desired.

Part B – Game Design

Create a new markdown file called `A1PartB.md`. You must use at least **two of: headings, external links, bullets, and bold/italics** within the markdown file.

1. Add the sources that you used for your game and any additional notes that you may have for your marker (including what version of Unity you are using).

2. Give your platformer a title and a story. This can be very simple. A couple sentences that establish: the world of the game, who the main character is, and the goal of the game.

3. Apply the Elemental Tetrad to analyze your platformer. Write 3 points about each of the four elements.